

Mapa de Lugares Turísticos de Quevedo

Aplicaciones Móviles - 6to Semestre

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Proyecto: GoogleMapsPrac

LINK DEL GITHUB:

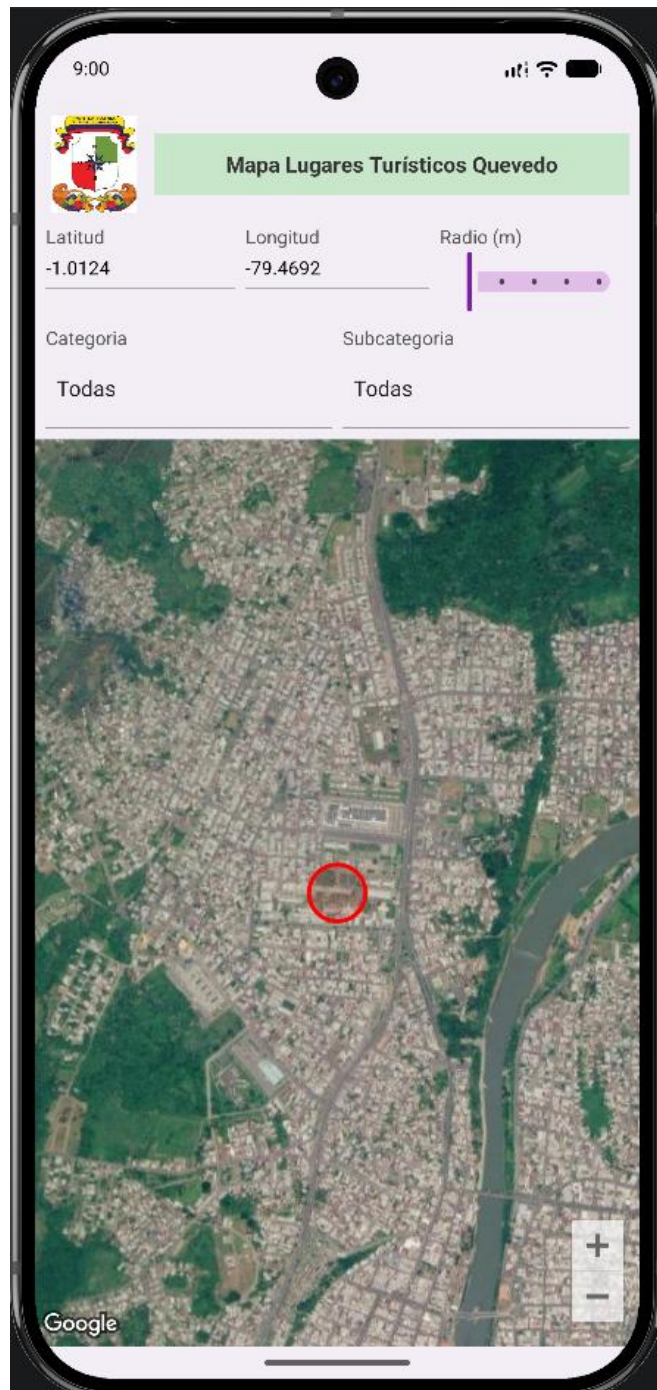
<https://github.com/Johnst2000/GoogleMapsPrac.git>

1. Capturas de la aplicación en funcionamiento

Pegar en cada espacio las capturas del emulador o dispositivo mientras la app está en ejecución.

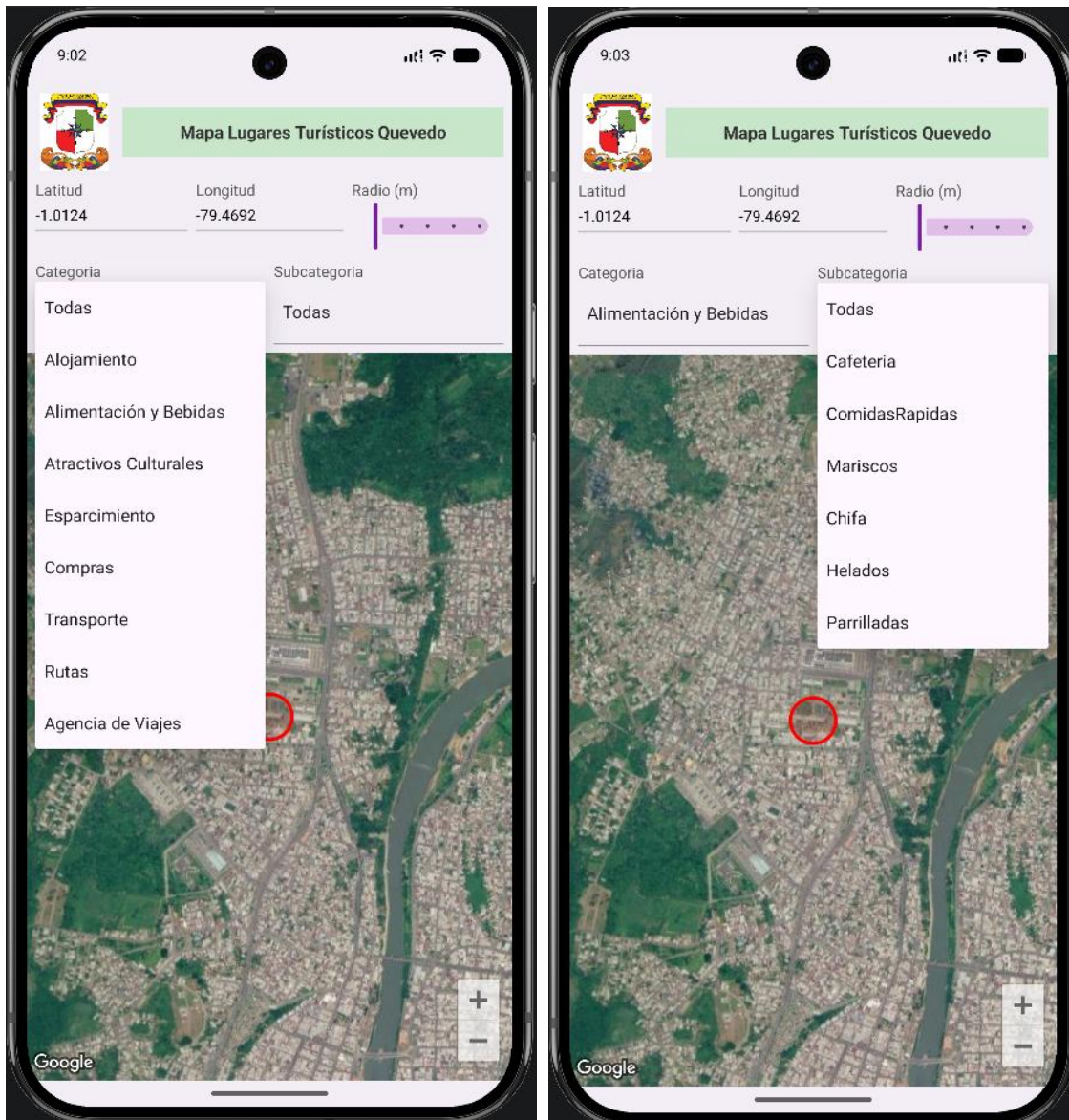
Captura 1 - Interfaz del mapa

Mapa en vista satélite, círculo rojo, slider de radio, latitud y longitud actualizadas al mover el mapa.



Captura 2 - Categorías y subcategorías

Spinners con la lista cargada desde la API. Mostrar una categoría seleccionada y sus subcategorías.



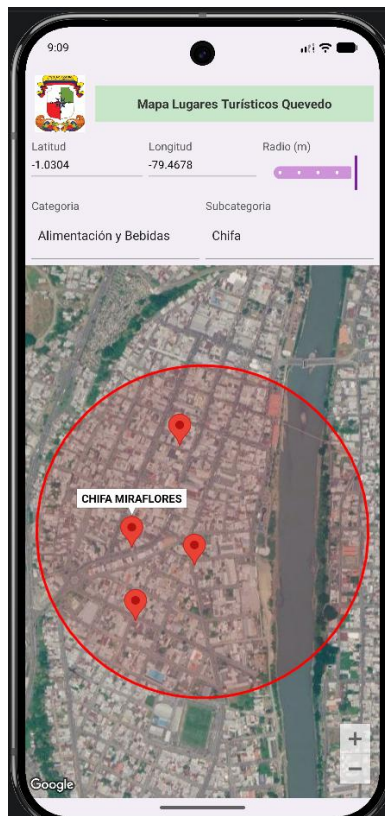
Captura 3 - Lugares turísticos en el mapa

Marcadores dentro del círculo con nombres visibles al tocar un pin.



Captura 4 - Filtros aplicados

Ejemplo: Categoría Alimentación y Bebidas + Subcategoría Chifas mostrando solo esos lugares.



2. Código fuente de la actividad principal

Actividad: MainActivity2.java (Java)

MainActivity2.java

```
package com.uteq.software.googlemaps;

import android.graphics.Color;
import android.location.Location;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.EditText;
import android.widget.Spinner;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.android.volley.Request;
import com.android.volley.RequestQueue;
import com.android.volley.toolbox.StringRequest;
import com.android.volley.toolbox.Volley;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.Circle;
import com.google.android.gms.maps.model.CircleOptions;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.Marker;
import com.google.android.gms.maps.model.MarkerOptions;
import com.google.android.material.slider.Slider;

import org.json.JSONArray;
import org.json.JSONException;
import org.json.JSONObject;

import java.util.ArrayList;

public class MainActivity2 extends AppCompatActivity implements OnMapReadyCallback {

    private static final String BASE_URL = "http://35.153.103.86/turismo10022025/";

    GoogleMap mapa;
    Double lat, lng;
    float radio;
    Circle circulo = null;
    Slider sliderRadio;
    EditText txtLatitud, txtLongitud;
    Spinner spinnerCategoria, spinnerSubcategoria;
    ArrayList<Marker> markers = new ArrayList<Marker>();
    RequestQueue requestQueue;
    ArrayList<String> idsCategoria = new ArrayList<>();
```

```

ArrayList<String> idsSubcategoria = new ArrayList<>();
String idCategoria = "";
String idSubcategoria = "";
boolean cargandoCategoria = false;
boolean cargandoSubcategoria = false;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main2);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });

    lat = -1.0123899038933037;
    lng = -79.46922232877745;
    radio = 1;

    txtLatitud = findViewById(R.id.txtLatitud);
    txtLongitud = findViewById(R.id.txtLongitud);
    sliderRadio = findViewById(R.id.sliderRadio);
    spinnerCategoria = findViewById(R.id.spinnerCategoria);
    spinnerSubcategoria = findViewById(R.id.spinnerSubcategoria);

    requestQueue = Volley.newRequestQueue(this);

    sliderRadio.addOnSliderTouchListener(new Slider.OnSliderTouchListener() {
        @Override
        public void onStartTrackingTouch(@NonNull Slider slider) {
            radio = slider.getValue();
            updateInterfaz();
        }

        @Override
        public void onStopTrackingTouch(@NonNull Slider slider) {
        }
    });

    spinnerCategoria.setOnItemClickListener(new AdapterView.OnItemClickListener() {
        @Override
        public void onItemClick(AdapterView<?> parent, View view, int position, long id)
    {
        if (cargandoCategoria) {
            return;
        }
        idCategoria = idsCategoria.get(position);
        cargarSubcategorias();
    }

    @Override
    public void onNothingSelected(AdapterView<?> parent) {
    }
    });

    spinnerSubcategoria.setOnItemClickListener(new AdapterView.OnItemClickListener()
    {
        @Override

```

```

        public void onItemSelected(AdapterView<?> parent, View view, int position, long id)
    {
        if (cargandoSubcategoria) {
            return;
        }
        idSubcategoria = idsSubcategoria.get(position);
        if (mapa != null) {
            updateInterfaz();
        }
    }

    @Override
    public void onNothingSelected(AdapterView<?> parent) {
    }
    });

    cargarCategorias();
    cargarSubcategorias();

    SupportMapFragment mapFragment = (SupportMapFragment)
        getSupportFragmentManager()
            .findFragmentById(R.id.map);
    mapFragment.getMapAsync(this);
}

@Override
public void onMapReady(GoogleMap googleMap) {
    mapa = googleMap;

    mapa.setMapType(GoogleMap.MAP_TYPE_SATELLITE);
    mapa.getUiSettings().setZoomControlsEnabled(true);

    mapa.moveCamera(CameraUpdateFactory.newLatLngZoom(new LatLng(lat, lng), 15));

    mapa.setOnCameraIdleListener(new GoogleMap.OnCameraIdleListener() {
        @Override
        public void onCameraIdle() {
            LatLng center = mapa.getCameraPosition().target;
            lat = center.latitude;
            lng = center.longitude;
            updateInterfaz();
        }
    });

    mapa.setOnMarkerClickListener(new GoogleMap.OnMarkerClickListener() {
        @Override
        public boolean onMarkerClick(Marker marker) {
            marker.showInfoWindow();
            return true;
        }
    });

    updateInterfaz();
}

private void cargarCategorias() {
    StringRequest request = new StringRequest(Request.Method.GET,
        BASE_URL + "categoria/getlistadoCB",
        result -> {
            try {

```

```

        ArrayList<String> nombres = new ArrayList<>();
        idsCategoria.clear();
        nombres.add(getString(R.string.todas));
        idsCategoria.add("");
        JSONArray lista = new JSONArray(result);
        for (int i = 0; i < lista.length(); i++) {
            JSONObject item = lista.getJSONObject(i);
            nombres.add(item.getString("descripcion"));
            idsCategoria.add(item.getString("id"));
        }
        cargandoCategoria = true;
        spinnerCategoria.setAdapter(new ArrayAdapter<>(
            MainActivity2.this,
            android.R.layout.simple_spinner_dropdown_item,
            nombres));
        cargandoCategoria = false;
    } catch (JSONException e) {
        e.printStackTrace();
    }
},
error -> {
});
requestQueue.add(request);
}

```

```

private void cargarSubcategorias() {
    idSubcategoria = "";
    if (idCategoria.isEmpty()) {
        ArrayList<String> nombres = new ArrayList<>();
        idsSubcategoria.clear();
        nombres.add(getString(R.string.todas));
        idsSubcategoria.add("");
        cargandoSubcategoria = true;
        spinnerSubcategoria.setAdapter(new ArrayAdapter<>(
            MainActivity2.this,
            android.R.layout.simple_spinner_dropdown_item,
            nombres));
        cargandoSubcategoria = false;
        if (mapa != null) {
            updateInterfaz();
        }
        return;
    }
}

```

```

StringRequest request = new StringRequest(Request.Method.GET,
    BASE_URL + "subcategoria/getlistadoCB/" + idCategoria,
    result -> {
        try {
            ArrayList<String> nombres = new ArrayList<>();
            idsSubcategoria.clear();
            nombres.add(getString(R.string.todas));
            idsSubcategoria.add("");
            JSONArray lista = new JSONArray(result);
            for (int i = 0; i < lista.length(); i++) {
                JSONObject item = lista.getJSONObject(i);
                nombres.add(item.getString("descripcion"));
                idsSubcategoria.add(item.getString("id"));
            }
            cargandoSubcategoria = true;
            spinnerSubcategoria.setAdapter(new ArrayAdapter<>(

```



```

        MainActivity2.this,
        android.R.layout.simple_spinner_dropdown_item,
        nombres));
        cargandoSubcategoria = false;
        if (mapa != null) {
            updateInterfaz();
        }
    } catch (JSONException e) {
        e.printStackTrace();
    }
},
error -> {
});
requestQueue.add(request);
}

private void updateInterfaz() {
    txtLatitud.setText(String.format("%.4f", lat));
    txtLongitud.setText(String.format("%.4f", lng));
    if (circulo != null) {
        circulo.remove();
        circulo = null;
    }
    CircleOptions circleOptions = new CircleOptions()
        .center(new LatLng(lat, lng))
        .radius(radius * 100)
        .strokeColor(Color.RED)
        .fillColor(Color.argb(50, 150, 50, 50));
    circulo = mapa.addCircle(circleOptions);

    String url = BASE_URL + "lugar_turistico/json_getlistadoMapa?lat=" + lat
        + "&lng=" + lng + "&radio=" + radius;
    if (!idCategoria.isEmpty()) {
        url += "&idc=" + idCategoria;
    }
    if (!idSubcategoria.isEmpty()) {
        url += "&idsc=" + idSubcategoria;
    }

    StringRequest request = new StringRequest(Request.Method.GET, url,
        this::processFinish,
        error -> {
        });
    requestQueue.add(request);
}

private boolean dentroDelCirculo(double markerLat, double markerLng) {
    float[] distancia = new float[1];
    Location.distanceBetween(lat, lng, markerLat, markerLng, distancia);
    return distancia[0] <= radius * 100;
}

private void processFinish(String result) {
    try {
        for (Marker marker : markers) {
            marker.remove();
        }
        markers.clear();

        JSONObject JSONObj = new JSONObject(result);

```

```

JSONArray jsonLista = JSONObj.getJSONArray("data");
for (int i = 0; i < jsonLista.length(); i++) {
    JSONObject lugar = jsonLista.getJSONObject(i);
    double markerLat = Double.parseDouble(lugar.getString("lat"));
    double markerLng = Double.parseDouble(lugar.getString("lng"));
    if (!dentroDelCirculo(markerLat, markerLng)) {
        continue;
    }
    markers.add(mapa.addMarker(
        new MarkerOptions().position(
            new LatLng(markerLat, markerLng)
        ).title(lugar.getString("nombre"))));
}
} catch (JSONException e) {
    e.printStackTrace();
}
}
}

```

3. Diseño en XML

Layout: activity_main2.xml

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/fondo_app"
    tools:context=".MainActivity">

    <LinearLayout
        android:id="@+id/headerMapa"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:padding="8dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <ImageView
            android:id="@+id/imgEscudo"
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:contentDescription="@string/escudo_quevedo"
            android:scaleType="fitCenter"
            android:src="@drawable/escudo" />

        <TextView
            android:id="@+id/txtTituloMapa"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_marginStart="8dp"
            android:layout_weight="1"
            android:background="@drawable/bg_titulo_mapa"
            android:gravity="center"
            android:padding="12dp"
            android:text="@string/titulo_mapa_turistico"
            android:textColor="@color/texto_titulo"
            android:textSize="16sp"
            android:textStyle="bold" />
    </LinearLayout>

    <LinearLayout
        android:id="@+id/panelControles"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:paddingStart="8dp"
        android:paddingEnd="8dp"
        android:paddingBottom="8dp">
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@id/headerMapa">
```

```
<LinearLayout
```

```
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:orientation="vertical"
    android:paddingEnd="4dp">
```

```
    <TextView
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/latitud"
        android:textColor="@color/label_texto"
        android:textSize="14sp" />
```

```
    <EditText
```

```
        android:id="@+id/txtLatitud"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@drawable/edittext_linea"
        android:focusable="false"
        android:inputType="numberDecimal|numberSigned"
        android:paddingTop="4dp"
        android:paddingBottom="6dp"
        android:textColor="@color/black"
        android:textSize="14sp" />
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:orientation="vertical"
    android:paddingStart="4dp"
    android:paddingEnd="4dp">
```

```
    <TextView
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/longitud"
        android:textColor="@color/label_texto"
        android:textSize="14sp" />
```

```
    <EditText
```

```
        android:id="@+id/txtLongitud"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@drawable/edittext_linea"
        android:focusable="false"
        android:inputType="numberDecimal|numberSigned"
        android:paddingTop="4dp"
        android:paddingBottom="6dp"
        android:textColor="@color/black"
        android:textSize="14sp" />
```

```
</LinearLayout>
```

```
<LinearLayout
```

```

        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:gravity="center_vertical"
        android:orientation="vertical"
        android:paddingStart="4dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="@string/radio"
            android:textColor="@color/label_texto"
            android:textSize="14sp" />

        <com.google.android.material.slider.Slider
            android:id="@+id/sliderRadio"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:stepSize="1"
            android:value="1"
            android:valueFrom="1"
            android:valueTo="5"
            app:thumbColor="#7B1FA2"
            app:trackColorActive="#CE93D8"
            app:trackColorInactive="#E1BEE7" />
    </LinearLayout>
</LinearLayout>

<LinearLayout
    android:id="@+id/panelFiltros"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:paddingStart="8dp"
    android:paddingEnd="8dp"
    android:paddingBottom="8dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/panelControles">

    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:orientation="vertical"
        android:paddingEnd="4dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="@string/categoria"
            android:textColor="@color/label_texto"
            android:textSize="14sp" />

        <Spinner
            android:id="@+id/spinnerCategoria"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:background="@drawable/edittext_linea"
            android:paddingTop="4dp"

```

```

        android:paddingBottom="6dp" />
</LinearLayout>

<LinearLayout
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:orientation="vertical"
    android:paddingStart="4dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/subcategoria"
        android:textColor="@color/label_texto"
        android:textSize="14sp" />

    <Spinner
        android:id="@+id/spinnerSubcategoria"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@drawable/edittext_linea"
        android:paddingTop="4dp"
        android:paddingBottom="6dp" />
    </LinearLayout>
</LinearLayout>

<fragment
    android:id="@+id/map"
    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="0dp"
    android:layout_height="0dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/panelFiltros"
    tools:context=".MainActivity2" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

4. AndroidManifest.xml

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.INTERNET"></uses-permission>

    <application

        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.GoogleMaps"
        android:usesCleartextTraffic="true">

        <activity
            android:name=".MainActivity"
            android:exported="true" />

        <activity
            android:name=".MainActivity2"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <meta-data android:name="com.google.android.geo.API_KEY"
            android:value="xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx"
        />
    </application>

</manifest>
```

5. Otras clases del proyecto

MainActivity.java: práctica anterior del silabo (polilínea y marcador UTEQ, páginas 16-18).

MainActivity.java

```
package com.uteq.software.googlemaps;

import android.graphics.Color;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.android.gms.maps.CameraUpdate;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.Marker;
import com.google.android.gms.maps.model.MarkerOptions;
import com.google.android.gms.maps.model.PolylineOptions;

public class MainActivity extends AppCompatActivity implements OnMapReadyCallback {
    GoogleMap mapa;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        SupportMapFragment mapFragment = (SupportMapFragment)
            getSupportFragmentManager()
                .findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);
    }

    @Override
    public void onMapReady(GoogleMap googleMap) {
        mapa = googleMap;

        mapa.setMapType(GoogleMap.MAP_TYPE_SATELLITE);
        mapa.getUiSettings().setZoomControlsEnabled(true);

        CameraUpdate camUpd1 =
            CameraUpdateFactory
                .newLatLngZoom(new LatLng(-1.0125791262991217, -79.46931397331672),
17);
        mapa.moveCamera(camUpd1);
    }
}
```



```

PolylineOptions lineas = new PolylineOptions()
    .add(new LatLng(-1.0120645693134682, -79.47161786876636))
    .add(new LatLng(-1.0123327483556732, -79.46734779188552))
    .add(new LatLng(-1.013502008720489, -79.46734779188552))
    .add(new LatLng(-1.0132123754584286, -79.47168224178466))
    .add(new LatLng(-1.0120645693134682, -79.47161786876636));
lineas.width(8);
lineas.color(Color.RED);
mapa.addPolyline(lineas);

LatLng punto = new LatLng(-1.0128261977353945, -79.46916096523441);
mapa.addMarker(new MarkerOptions()
    .position(punto)
    .title("UTEQ"));

mapa.setOnMarkerClickListener(new GoogleMap.OnMarkerClickListener() {
    @Override
    public boolean onMarkerClick(Marker marker) {
        marker.showInfoWindow();
        return true;
    }
});
}
}

```